Installing and operating instructions

DRELLOSCOP® 255 strobe unit

Read these instructions carefully **before** you attempt to connect or operate this equipment and follow storage instructions.



DRELLO

The installing and operating instructions, (version 1.0255.2B002e) are for the DRELLOSCOP 255 strobe unit

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Introduction

The installing instructions for the DRELLOSCOP 255 are written for technical personnel (electricians and maintainers) who need to install and maintain the unit. Technical personnel must know and follow all applicable electrical regulations.

The operating instructions are written for workers who need to adjust and operate the unit for manufacturing and control purposes.

Read these instructions carefully to learn how to work safely with your DRELLOSCOP.

Intended use

The DRELLOSCOP 255 is a professional strobe unit designed for visual quality control of operating processes.

No other use of the unit is authorised; any risks are borne by the user alone.

Never allow anyone other than a properly qualified electrical technician to install or maintain the unit.

For any questions concerning operation of the unit or special types of application that are not addressed in the instructions, contact:

DRELLO GmbH & Co. KG	Telephone	+49-(0)2161-909-6
Max-Reger-Straße 35a	Fax	+49-(0)2161-909-700
D-41179 Mönchengladbach	E-mail	info@drello.de

For more information about DRELLO GmbH & Co. KG and their range of products, visit the web site at **http://www.DRELLO.de**.

Nameplate

Your unit is identified by a unique serial number, which you will find printed on the nameplate on the back of the DRELLOSCOP 255.

You should record the serial number, identified by SN (see illustration), and quote that number whenever calling for technical assistance or to order spare parts.



Electromagnetic compatibility

The DRELLOSCOP 255 complies with the requirements of the European Electromagnetic Compatibility Directive (89/336/EEC).

Safety

Electrical hazards

- Never attempt to operate your DRELLOSCOP 255 unit and associated equipment from any power source other than the one recommended herein.
- Never allow anyone other than a properly qualified electrical technician to install or maintain the unit.
- Immediately disconnect your unit from the power source and from all other equipment if you have any reason to suspect malfunction (e.g. if you detect smoke or an acrid smell in the air).
- Residual high voltages may be present inside the unit even when it is disconnected from power! The housing should only be opened by a suitably qualified electrical technician, for the purpose of replacing the fuses and the flash tube. There are no other user-maintainable parts in the DRELLOSCOP.
- Protect this equipment against all fluid spills (e.g. from coffee cups placed on top of the unit, cleaning fluids, etc.). In the event of a spill, immediately stop using the unit.
- If you intend to leave your DRELLOSCOP 255 unused for a protracted period, disconnect it from the power source and ensure it is stored in a dry location where it will be protected from weather and outside influences.
- Treat all electrical cables with care to avoid any damage from crushing, pinching etc. Replace all damaged cables immediately.
- If you suspect a dangerous condition, disconnect the unit from the power source and lock it out to prevent inadvertent start-up.
- Examples of a dangerous condition include (but are not limited to):
  – visible damage to the unit or the power cable,
  – the unit is not operating correctly,
  – the interior of the unit was exposed to rain or other source of moisture,
  – condensation has formed on the unit,
  – foreign objects have penetrated inside the unit,
  – the allowed storage/operation temperature has been fell below or exceeded

Hazards associated with strobe operation

Strobe operation at frequencies between 10....20 Hz has been known to induce epileptic seizures in susceptible persons. The likelihood of a seizure grows with increasing contrast between strobe brightness and ambient light intensity. The hazard is not limited to diagnosed epilepsy sufferers. A seizure can lead to injuries (e.g. risk of falling).

1) Do not adjust the flash frequency higher than it is necessary to obtain an image of high contrast.
2) Wherever possible, use natural ambient light
3) Make sure that no persons diagnosed with epilepsy are in the vicinity of the strobe, while you are operating it.

In case your DRELLOSCOP 255 has been properly set up, rotating machine parts look as they are standing still. Persons who are not informed about this, are in danger of touching the rotating machine or to be caught by it. Those persons may be severely injured or killed.

- Inform all persons who for any reason may find themselves in the area illuminated by the DRELLOSCOP 255 about the strobe effect and the associated danger arising from the optical illusion.

- Restrict access to the strobe-illuminated area to those persons who have a legitimate need to enter it.

Installation

The installation work described in these instructions should be entrusted to suitably trained and qualified technicians. Applicable safety regulations and technical standards and codes must be complied with at all times.

Operating the unit safely

Leave sufficient clearance on all sides of your DRELLOSCOP unit to allow free circulation of air around it. This is to prevent the unit overheating and becoming seriously damaged.

Safe repairs

Do not attempt to perform any repair work on the unit, to avoid injury or further damages to the unit or to other property. If your unit is defective, contact the customer service representative of DRELLO GmbH & Co. KG.

Safe maintenance

The maintenance work described should be entrusted to suitably trained and qualified technicians. Applicable safety regulations and technical standards and codes must be complied with at all times.

You should not attempt to perform any maintenance other than that described in these instructions, to avoid injury or further damages to the unit or to other property.

Symbols used

In these instructions you will see two types of safety instruction:



Danger!
Indicates an imminent risk. Failure to follow the instructions can lead to serious injury or death.



Caution!
Indicates a risk of damage to the unit or other equipment if the instructions are not followed.

Useful hints and tips on using the DRELLOSCOP 255 are marked like this:



Note
Gives instructions on how best to use the unit.

Product description

Principle of operation

Your DRELLOSCOP 255 is a flashlight stroboscope. It transfers pulses of an external pulse trigger unit to the built-in flash tube. The light flashes produced like that are transferred via the connected light conductor, which illuminates then the required moving object.

If the frequency of the light flashes matches the frequency of rotation or oscillation of the moving object, the illusion is created that the object is stationary.

The flash energy can be adapted to the requirements of the vicinity, the camera and the controlling person.

Triggering flash frequency by external pulses

In external strobe trigger mode a trigger unit emits the pulses for the flash frequency. In this mode the flash frequency follows automatically the speed of the object. In case of changing speed, flash frequency changes automatically, so that the stationary image is maintained.

List of equipment

1	Flashlight Stroboscope DRELLOSCOP 255 – 01 or DRELLOSCOP 255 - 02	part no. 1.0255.60001 part no. 1.0255.60002
	for 90...260V power supply incl. flash tube	
1	6-pole cable plug for DRELLOSCOP 255 – 01	part no. 17.01.05
1	3-pole cable socket for DRELLOSCOP 255 – 02	part no. 17.01.04
1	installing and operating instructions	part no. 1.0255.2B002e

Standard accessories

▪	Quartz-glass flash tube, type 804-1	part no. 1.B804.20001

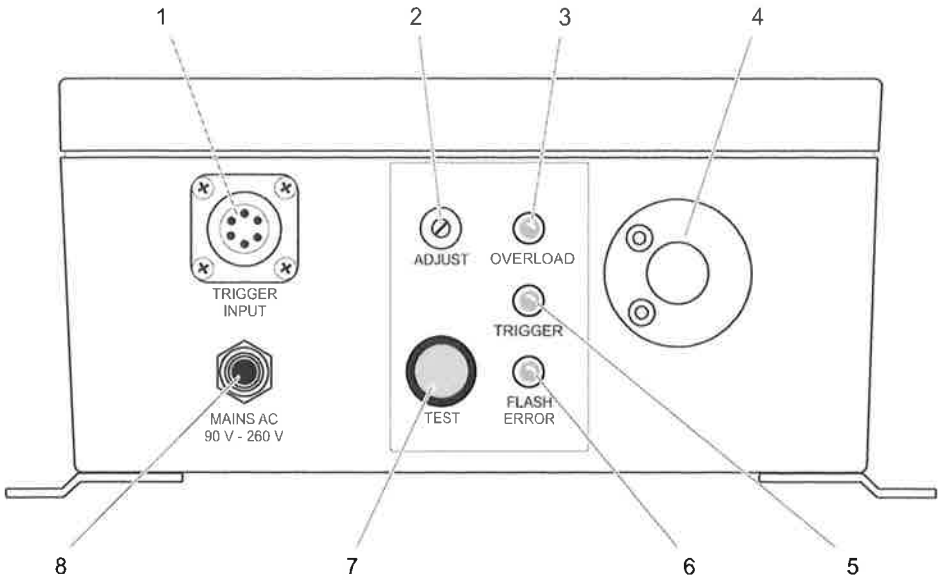
Recommended accessories

▪	glass fibre light conductor GL 1,0/13 length 1 m, active diameter 13 mm	part no. 19.13.18

Light conductors according to customers specification can be supplied on demand.

Controls and connections

Front panel DRELLOSCOP 255 - 01



1 **6-pole female socket *TRIGGER INPUT***
for connecting a trigger unit

2 **turning knob or adjusting screw *ADJUST***
for manual adjustment of flash energy

3 **LED *OVERLOAD***
at overload it changes from green (permissible range) to red (overload)

4 **Connection socket**
for feeding the light conductor

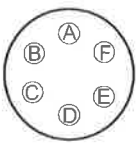
5 **LED *TRIGGER***
flashes in timing of trigger unit

6 **LED *FLASH ERROR***
for indication of faults in flash frequency

7 **Key *TEST***
for releasing flashes without trigger pulse

8 **power cable**

Assignment trigger pulse *TRIGGER INPUT*

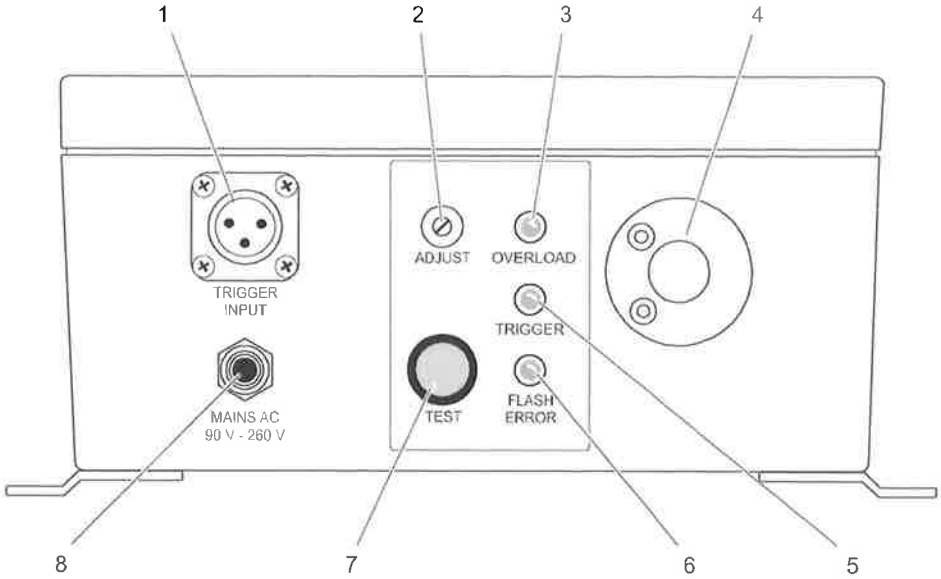


Connection	Assignment	Direction	Level
A	N. C.		
B	Supply voltage	Output	+5 V DC, max. 30 mA
C	Reference potential +5 V DC		0 V
D	Earthing		PE
E	Reference potential trigger input	Input	0 V
F	Trigger input	Input	3 ... 12 V_s

Note
Pin *C* (0 V) is connected galvanic coupled with the protective wire

Controls and connections

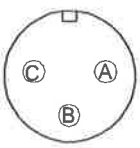
Front panel DRELLOSCOP 255 - 02



1. **3-pole female socket *TRIGGER INPUT***
   for connecting a trigger unit
2. **Turning knob or adjusting screw *ADJUST***
   for manual adjustment of flash energy
3. **LED *OVERLOAD***
   at overload it changes from green (permissible load) to red (overload)
4. **Connection socket**
   for feeding the light conductor
5. **LED *TRIGGER***
   flashes in timing of trigger unit
6. **LED *FLASH ERROR***
   for indication of faults in flash frequency
7. **Key *TEST***
   for releasing flashes without trigger pulse
8. **Power cable**

Assignment Pulse Trigger *TRIGGER INPUT*

Connection	Assignment	Direction	Level
A	Reference potential pulse input	Input	0V
B	Pulse input	Input	3...12Vs
C	Earthing		PE



Note
Pin *A* (0 V) and Pin *C* (0 V) are galvanic coupled with the earthing.

Installing and connecting your unit

Preparations



Caution!
Choose a location where the unit will be protected against extreme temperatures and high humidity. Remember that it must not be exposed to temperatures lower than 0 °C or higher than 40 °C.

The mains electrical voltage must be the same as the supply voltage specified in the nameplate.

Do not install your DRELLOSCOP in the vicinity of any heaters.

Avoid installing it where it will be exposed to direct sunlight.

Ensure that air can circulate freely around the unit.

Make sure that your DRELLOSCOP is not exposed to vapour or liquids.

The lighting should not be excessively bright to get the best performance out of your DRELLOSCOP.

Ensure the unit is safe from vibration and shock effects.

Installation

The DRELLOSCOP can be used as desk-type or you can mount it on the wall. Maximum distance to the control area depends on the lengths of the used light conductor. It is important that the light conductor can be positioned without problems at the place of operation and may not be bent.

You can also mount the DRELLOSCOP on the wall. For this purpose take 6 mm screws and suited dowels in order to screw the unit at the four mounting brackets on to the wall.



Note
The screws for mounting do not belong to the delivery scope.

Connecting peripheral equipment



Electrical shock hazard!
Disconnect electrical power to all equipment before making any connections.



Electrical shock hazard!
Take care after connection that all cables are laid perfectly not creased, crushed or damaged elsewhere.

- Install external trigger unit (if used), following the manufacturer's instructions for installation and operation.
- Plug in the cable of the trigger unit in the TRIGGER/INPUT of the unit and secure it.
- Connect the power cable to the mains.

The DRELLOSCOP 255 is ready for operation now.

Powering the unit on and off

This unit is part of a system and cannot be powered on or off alone.

Verify that all connections are correctly made and secure prior to powering the system on.

- Switch on the power supply to the system.
  The unit is powered on.
- In case of emergency interrupt the power supply to the system or plug off the plug of the DRELLOSCOP out of the mains socket.

Using the unit

Verify that all peripheral equipment connections are correctly made and secure prior to using the unit.

Ensure your DRELLOSCOP 255 unit and all peripheral equipment are switched on.

The DRELLOSCOP 255 is ready for operation. The LED TRIGGER is flashing in the same cycle as pulse trigger.

Controlling strobe function

You can also check function of the DRELLOSCOP 255 if it is ready for operation, but does not yet receive any signals from the pulse trigger. For this:

- Push key TEST
  A flash is released

As soon as trigger unit provides pulse to the DRELLOSCOP 255, LED TRIGGER starts flashing in same cycle as pulses.

Adjusting light conductor

In case the light conductor is positioned best possible, you can operate your DRELLOSCOP with the lowest possible flash energy. That saves the unit, flash tube and also the eyes of the persons working in the surrounding.

To adjust the position of the light conductor for maximum light output, act as follows:

- Loosen the set screw with the hexagon socket head screw M3 on the upper right side of the connector socket for the light conductor.
- Shift the light conductor in the mounting till you have found the optimum position for the maximum light output.
- Now fasten again the set screw with the hexagon socket head screw M3.

Adjusting flash energy

Before adjusting the flash energy make sure that the light conductor is positioned best possible and that the ambient light in not too bright.

For adjusting the flash energy (ADJUST) you will find on the front side of the DRELLOSCOP either a potentiometer knob or a set screw, which has to be turned by means of a slotted-type screwdriver.

- Shield the ambient light best possible to keep the flash energy as low as possible..
- Turn the rotary knob ADJUST counter-clockwise to the stop in order to start with the lowest possible flash energy.
- Now slowly turn the rotary knob clockwise till you have achieved the optimum flash energy.



Note
Does the LED OVERLOAD lights up red, reduce the flash energy till it lights up green.

Maintenance

The maintenance works described in this manual are only to be done by skilled qualified personnel. In any case technical personnel must know and follow all applicable electrical regulations.

Except the regular replacement of quartz-glass flash tubes the DRELLOSCOP 255 is maintenance-free.

Check the cabling for safety and tight seat before each starting-up of the system. Damaged cable has to be exchanged.

Troubleshooting – Causes and Solutions

This chapter describes troubleshooting at the DRELLOSCOP 255, possible causes and solutions. Do not do any repair works which are not described in this chapter. In case of all other troubleshooting contact always DRELLO GmbH & Co. KG.

Troubleshooting	Possible cause	Solution
Flash tube does not flash. The LED FLASH ERROR lights up.	Flash tube is worn or defective	Replace flash tube, see chapter "Replace flash tube"
Flash tube flashes irregularly or does not flash at all, although LED FLASH ERROR does not light up	Control pulses are faulty.	Check control connector. Push red key TEST to find out if function of the DRELLOSCOP are perfect. If the flash tube flashes, you can continue operating your DRELLOSCOP . Does the flash tube not flash, replace flash tube (see last chapter).
LED OVERLOAD lights up	Flash energy is adjusted too high	Reduce Flash energy, see chapter "Adjusting flash energy
Unit does not work. All LEDs are dark	Power supply is interrupted	Check power supply
	One of the fuses is defective	Replace fuses in the interior of the unit, see chapter "Replacing fuses"

Note
The LED FLASH ERROR indicates, that the flash tube is tending to flash interruptions and has to be replaced.

This LED does not have any influence on the functions of the unit. By actuating the key TEST or by switching off and re-start of the system the DRELLOSCOP can be re-set.

Replacing flash tube

The flash tube of the DRELLOSCOP 255 is located in the interior of the unit. For replacing the flash tube always wear cotton gloves.

The housing may only be opened by a qualified electrical technician and only for the purpose of replacing the fuses or the flash tube.



Electrical shock hazard!
The flash tube must be replaced by a qualified electrical technician. Follow safe electrical practice at all times.



Electrical shock hazard!
Ensure that the unit is disconnected from electrical power and all other electrical equipment and secure it against re-start.



Electrical shock hazard!
Capacitors inside the unit are capable of causing potentially deadly electrical shocks. Before opening the unit, disconnect it from the power supply and wait at least 5 minutes to allow dangerous charges to dissipate.

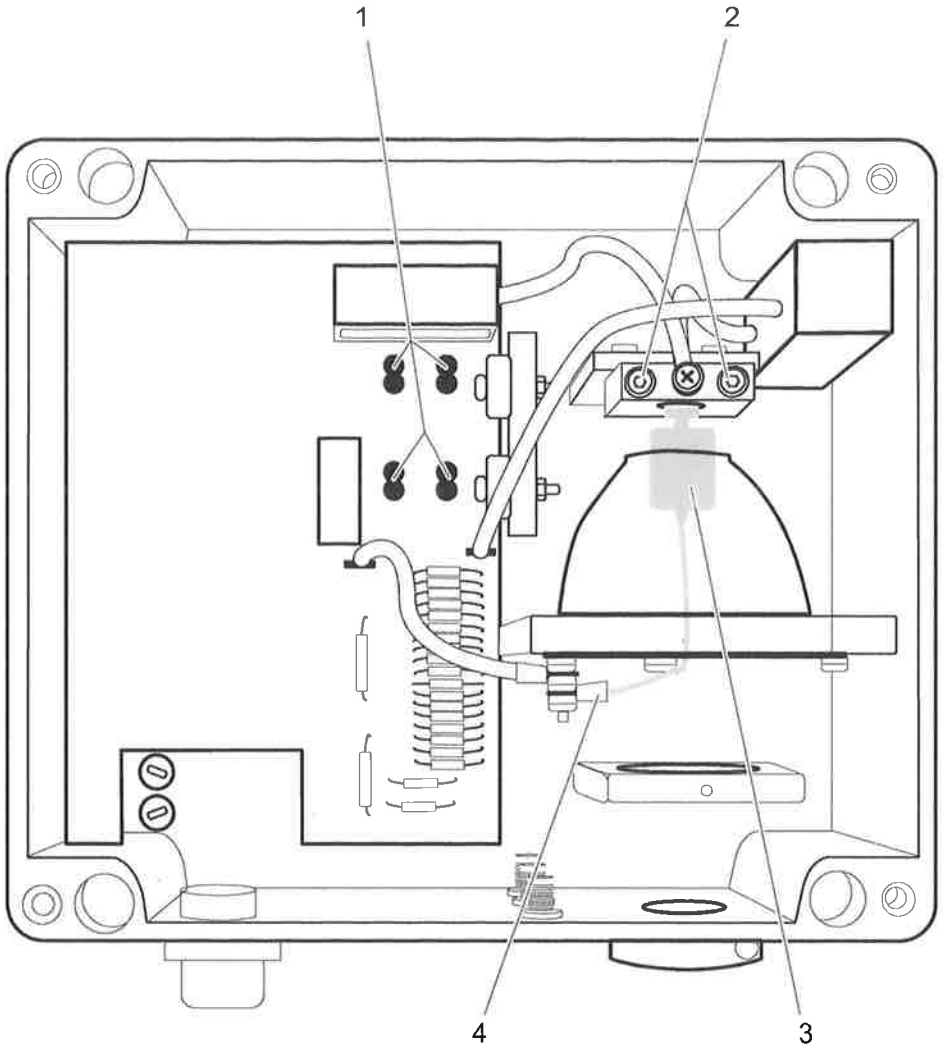


Burning hazard
During operation the unit and especially the flash tube become extremely hot. Wait at least 5 minutes to let the unit and the flash tube cool down.



Caution!
Before replacing the flash tube you must ensure for sufficient earthing to avoid static discharges. In the interior of the unit there are electronic components which can be destroyed by static discharges.

- Disconnect DRELLOSCOP 255 from power supply and wait at least 5 minutes.
- Make sure that the unit is cooled down.
- If necessary, remove the unit from the mounting brackets and place it on its bottom side.
- Loosen the four screws at the housing cover with a cross-tip screwdriver and carefully put the cover left beside the unit without damaging the cable which is fastened at the housing cover. Short-circuit the contacts of the flash capacitor 1 with a test and earth lead (test voltage 3000 V) to make sure that the flash capacitor is totally discharged.
- Unscrew earth lead 4 at the reflector with a 7 mm key. For this take away the M4-nut, lock washer and washer. Pull off the cable lug from the earth connection.
- Loosen the set screws on the left and right side of the tube holding 2. Never loosen the electrical connection in the middle.
- Pull in forward direction flash tube 3 out of the reflector.
- Insert new flash tube and press it against the stop.
- Fasten set screws of set block 2.
- Put cable lug of the flash tube onto the earthing lead 4. Pull on washer, lock washer and M4-nut and fasten with a 7 mm key.
- Place housing cover on the housing and fasten it
- If necessary mount the unit into the mounting brackets and locate it to its operation place.
- Plug in unit to power supply
- Push key TEST to check function of the flash tube.



Replacing fuses

The DRELLOSCOP 255 has two fuses. In order to replace them, the unit has to be opened. The housing may only be opened by a qualified electrical technician and only for the purpose of replacing the fuses or the flash tube.

The fuses are accessible through a clearance in the upper PCB. They are located beside each other in the interior of the unit on the lower PCB.

Electrical shock hazard!
All fuses of the unit must be replaced by a qualified electrical technician. Follow safe electrical practice at all times.

Electrical shock hazard!
Ensure that the unit is disconnected from electrical power and all other electrical equipment and secure it against re-start.

Electrical shock hazard!
Capacitors inside the unit are capable of causing potentially deadly electrical shocks. Before opening the unit, disconnect it from the power supply and wait at least 5 minutes to allow dangerous charges to dissipate.

Caution!
Before replacing the flash tube you must ensure for sufficient earthing to avoid static discharges. In the interior of the unit there are electronic components which can be destroyed by static discharges.

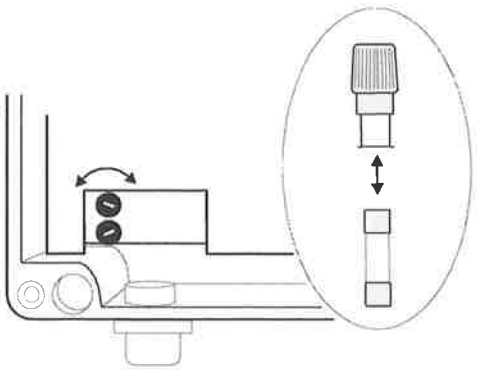
Caution!
Before inserting a new fuse check what was the cause for the defect

Caution!
Only use a fuse with the values: T 1,0 A/250 V, Size 5 × 20 mm

- Disconnect DRELLOSCOP 255 from power supply and wait at least 5 minutes. Make sure that the unit is cooled down .
- If necessary, remove the unit from the brackets and put it on its bottom side. The operation panel is turned towards you now.
- Unscrew the four screws of the housing cover with a cross-tip screwdriver and carefully put the cover left beside the unit without damaging the cable which is fastened at the housing cover.
- Short-circuit the contacts of the flash capacitor 1 with a test and earth lead (test voltage 3000 V) to make sure that the flash capacitor is totally discharged.

The two fuses are located on the left side of the lower PCB and are accessible through a clearance in the upper PCB.

- Turn the cover of the plugged fuse base counter-clockwise for approx. 45° and pull it out.
  Take away the defective fuse.
- Insert the new fuse into the cover and carefully push both to the stop into the *plugged fuse base*.
- Turn it clockwise 45° with slight pressure and fix it.
- The fuse is properly fixed if you cannot pull it out of this position.
- Put the cover on the housing again and screw it.
- If necessary mount the unit into the mounting brackets and locate it to the place of operation.
- Plug in unit to power supply.
- Push key TEST to check function of the unit.



Annex

Repairing defects

If your DRELLOSCOP 255 requires repairs, contact:

DRELLO GmbH & Co. KG

Customer Service

Max-Reger-Straße 35a
D-41179 Mönchengladbach
Germany

Phone +49-(0)-2161-909-6

Fax +49-(0)-2161-909-700

E-mail service@drello.de

Technical specifications

Supply voltage	90...260V AC 40-60 Hz %), 40...60 Hz
Secondary voltage:	5 V DC, max 30 mA, only at 6-pole jack
Power consumption	max. 50 W approx. 7 W on no load
Fuse rating interior	2 x T1A, 250 V (5x20 mm)
Strobe trigger connection	1 × 6-pin female socket for DRELLOSCOP 255-01 or 1 x 3-pole male socket for DRELLOSCOP 255-02
Strobe trigger	Pulses +3V...12V 30µs...1ms period
Flash frequency	0 ... 50 Hz depending on the power adjusted
Flash energy:	max. 2.5 Ws/flash
Flash duration	5...8µs (half width of flash) spectral range 300...1100 mm
Flash tube:	type 804-1
Ambient conditions	temperature: 0 … +40 °C
Storage conditions	temperature: -20 … +60 °C store in dry place
Housing	light metal die casting
Protection class	IP 54 (at plugged in light conductor and connected trigger cable
Dimensions (W × H × D)	230 × 120 × 200 mm
Weight	approx. 4.3 kgs
Insulation rating class	complies with IEC 348 and VDE 0411
Standards:	VDE 0110b, VDE 0411, VDE 05501 IEC 348 EN 50082-2, EN 55011B, EN 55140, EN 55141, EN 61000-4-2, EN 61000-4-4